

Walmart Store Sales Forecasting Process Flow Documentation

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Process Flow Documentation

Walmart Store Sales Forecasting

1. Business Understanding

Walmart operates multiple retail stores across different regions, and weekly sales are influenced by a combination of store-level characteristics, seasonal patterns, holidays, promotional markdowns, and macroeconomic indicators such as fuel price, consumer price index (CPI), and unemployment.

The objective of this project is to forecast **weekly sales (`Weekly_Sales`)** for Walmart stores using historical sales records, store metadata, and economic/promotional features. Accurate forecasting supports better inventory planning, staffing, and promotional decision-making.

Business Impact: [TO BE FILLED]

2. Data Collection

Three datasets were used for this project:

Dataset	Description
<code>train.csv</code>	Historical weekly sales data by Store and Department
<code>features.csv</code>	Store-level economic and promotional indicators
<code>stores.csv</code>	Store metadata (Type, Size)

Data source: [TO BE FILLED]

3. Data Integration

The datasets were merged into a single unified dataset for modeling:

1. `train.csv` was merged with `features.csv` on the common keys:
 - `Store`
 - `Date`
 - `IsHoliday`
2. The resulting dataset was then merged with `stores.csv` on:
 - `Store`

Final integrated dataset size: 421,570 rows

4. Data Understanding

After integration, the dataset contained the following features:

- `Store`
- `Dept`
- `Date`
- `IsHoliday`
- `Temperature`
- `Fuel_Price`
- `MarkDown1`
- `MarkDown2`
- `MarkDown3`
- `MarkDown4`
- `MarkDown5`
- `CPI`
- `Unemployment`
- `Type`
- `Size`

Target Variable: `Weekly_Sales`

Initial exploration was carried out using summary statistics (`describe()`) to understand the distribution, scale, and range of numerical features.

5. Data Cleaning

Data cleaning steps performed: [TO BE FILLED]

(No explicit missing-value handling, imputation strategy, or duplicate-removal steps were provided. This section should be completed with the specific cleaning operations applied, e.g., handling of missing `MarkDown` values, date parsing, dtype conversions, etc.)

6. Exploratory Data Analysis (EDA)

The following EDA steps were performed:

- Summary statistics using `describe()`
- Weekly Sales distribution plot
- Outlier analysis using boxplots
- Correlation heatmap

Key EDA Findings:

- Weekly Sales distribution is highly right-skewed
 - Weekly Sales contains large outliers
 - `MarkDown` variables contain significant outliers
 - Store sizes vary substantially
 - Economic variables (`CPI`, `Fuel_Price`, `Unemployment`) vary over time
-

7. Feature Engineering

Date-based features derived from Date :

- Year
- Month
- Week
- Quarter
- Month_End

Encoding:

One-Hot Encoding was applied to the Type (Store Type) column, generating:

- Type_B
- Type_C

(Type_A was used as the reference/baseline category.)

8. Model Development

Three models were built and evaluated:

Model 1: Linear Regression

Metric	Value
MAE	11,908
RMSE	18,527
R ²	0.052

Observation: Linear Regression performed poorly and could not capture the non-linear relationships present in the data.

Model 2: Random Forest Regressor

Status: [TO BE FILLED]

Model 3: XGBoost Regressor

Hyperparameter Tuning: RandomizedSearchCV

Best Parameters:

```
n_estimators: 700
max_depth: 4
learning_rate: 0.05
min_child_weight: 5
colsample_bytree: 0.9
```

Results:

Metric	Value
MAE	5,587
RMSE	10,009
R ²	0.723

Top Important Features (XGBoost):

- Dept

2. Type_B
3. Size
4. Store
5. CPI
6. Week
7. Markdown3
8. Unemployment
9. Month
10. Type_C

9. Model Comparison

Model	MAE	RMSE	R ²
Linear Regression	11,908	18,527	0.052
Random Forest Regressor	[TO BE FILLED]	[TO BE FILLED]	[TO BE FILLED]
XGBoost Regressor	5,587	10,009	0.723

Based on the available results, **XGBoost Regressor** significantly outperformed Linear Regression across all evaluation metrics.

10. Final Model Selection

Selected Model: XGBoost Regressor

Justification: Among the models with completed evaluation, XGBoost Regressor achieved the lowest MAE (5,587), lowest RMSE (10,009), and highest R² (0.723), substantially outperforming Linear Regression. Final selection pending completion of Random Forest Regressor evaluation. [TO BE FILLED — update if Random Forest results become available and outperform XGBoost]

11. Conclusion

This project demonstrates an end-to-end sales forecasting pipeline — from data integration across three sources, through EDA and feature engineering, to model development and comparison. The XGBoost Regressor, tuned via RandomizedSearchCV, provided the strongest performance among the models evaluated with complete results (MAE = 5,587, RMSE = 10,009, R² = 0.723), compared to a baseline Linear Regression model which was unable to capture the non-linear structure in the data.

Future Work / Next Steps: [TO BE FILLED]

Document generated based solely on information provided by the project author. Sections marked [TO BE FILLED] require additional input before this documentation is considered complete.